

Arya Hosseini

Montreal, QC, Canada

[Portfolio Website](#)

[GitHub Repository](#)

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SUMMARY

Junior Game Programmer with professional and indie experience building games in **Unity** and **Unreal Engine**. Skilled in **C#** and **C++**, with a strong foundation in gameplay programming, systems design, and architecture. Seeking opportunities to build high-quality, interactive games while collaborating with passionate development teams.

SKILLS

Engines & Tools: Unity, Unreal Engine, SFML, ImGui, Git, Shader Graph, Maya

Programming Languages: C#, C++ , Lua, SQL, JavaScript, HTML/CSS

Languages: English, French

PROFESSIONAL EXPERIENCE

Unity Gameplay Programmer Intern - Baobab Games

May 2024 - Aug 2024

DION – Multiplayer Survival

Montreal.Canada

- Expanded the **construction system** by extending existing tools and integrating **18 new modular parts** with unique constraints, enabling rapid content iteration without engineering support.
- Designed and implemented an **interaction framework** using **interfaces**, allowing **dynamic interactions** with world objects.
- Developed a **network-synced storage system** using **Mirror Networking**, integrating it with **UI** and **inventory systems**; optimized item spawning with **object pooling** for smooth, synchronized multiplayer gameplay.
- Developed custom **Unity editor tools** to streamline prefab generation and level design workflows, **improving team efficiency** and **reducing repetitive setup time**.
- Collaborated with **artists** and **QA** in an **Agile workflow**, addressing feedback and ensuring efficient cross-team iteration.
- Contributed to **design discussions** and **prototype** reviews to improve **user experience** and **gameplay balance**.

PERSONAL PROJECT

Hallowfall - Indie Project

2024 - Present

Action Roguelike - Unity (Solo Project)

[View Project](#)

- Designed a **state-driven player architecture** using the **State Pattern**, separating movement, input, combat, and animation for clean extensibility.
- Developed a **procedural world generation system** using zone-based architecture and **binary space partitioning**; optimized with **coroutine batching** and **zone culling**.
- Built a **data-driven ability system** with **ScriptableObjects** and **interfaces**, supporting **randomized upgrades**, **composable abilities**, and seamless **UI integration**.
- Implemented a **custom Flow Field Pathfinding system**, replacing Unity's NavMesh to support **dynamic procedural zones** and **efficient multi-agent navigation**.
- Created a **modular Enemy AI framework** based on **FSMs** and **ScriptableObject** defined behaviors, supporting **reusable states** and **scalable** boss designs.
- Optimized **CPU** and **GPU performance** using the **Unity Profiler**, reducing overhead through code refactoring, memory management and render batching.

DEC – Computer Science, Video Game Programming

2022 - 2024

LaSalle College

Montreal.Canada

Bachelor of Engineering Science (1 year completed)

2021 - 2022

University of Tehran

Tehran.Iran